
Homework 1

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MA 542 Section A1

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Problem 1

The set $\{(1, 0), (0, 1)\}$ forms a basis for $\mathbb{R}^2$ as an $\mathbb{R}$ -vector space. To prove this, let $(x, y) \in \mathbb{R} \times \mathbb{R}$. Since $x, y \in \mathbb{R}$,

$$(x, y) = x(1, 0) + y(0, 1).$$

Also note that the two elements are trivially linearly independent, so they form a basis. Consider the map $\phi : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{C}$ defined by

$$(a, b) \mapsto a + bi.$$

Let $r \in \mathbb{R}$ and $(c, d) \in \mathbb{R} \times \mathbb{R}$, and note that

$$\phi(r(a, b)) = r(a + bi) = r\phi(a, b),$$

$$\phi((a, b) + (c, d)) = \phi(a + c, b + d) = (a + c) + (b + d)i = (a + bi) + (c + di) = \phi(a, b) + \phi(c, d).$$

Therefore, ϕ is an $\mathbb{R}$ -linear map, and it also gives a bijection of basis elements:

$$\phi(1, 0) = 1, \quad \phi(0, 1) = i.$$

Therefore, ϕ is a vector space isomorphism, and $\mathbb{R} \times \mathbb{R} \cong \mathbb{C}$.

Problem 2

No to both. Note that $(1, 0), (0, 1) \neq 0 := (0, 0)$. Also note that

$$(1, 0) \cdot (0, 1) = (1 \cdot 0, 0 \cdot 1) = (0, 0).$$

Therefore, $\mathbb{R} \times \mathbb{R}$ has zero divisors, and is not an integral domain nor a field.

Problem 3

No. The number i must satisfy $i^2 = -1$. In $\mathbb{R} \times \mathbb{R}$, -1 is simply the additive inverse of 1, and $1 = (1, 1)$ fairly obviously:

$$(a, b)(1, 1) = (a1, b1) = (a, b).$$

We thus seek an element $(a, b) \in \mathbb{R}^2$ such that $(a, b)^2 = (-1, -1)$. Since multiplication is defined elementwise, it follows that $a^2, b^2 = -1$, with $a, b \in \mathbb{R}$. This is a contradiction because there exists no $a \in \mathbb{R}$ such that $a^2 = -1$.